UNIT 4 · 8 hours

Routh–Hurwitz Stability Criterion



Stability of Systems

Control Systems (22EEG53)
Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

- Construct the Routh array
- State the necessary and sufficient condition
- Handle a zero in the first column and a row of zeros
- Find the range of K for stability and the marginal frequency

Syllabus context

Stability of systems: characteristic equation, location of roots, absolute and relative stability, Routh–Hurwitz criterion (necessary and sufficient conditions), root locus concept, rules for construction of root locus.

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Constructing the Routh array

1

Write polynomial

$a0s^n + a1s^{n-1} + a2s^{n-2} + \ldots$ (a0 > 0).

2

First two rows

Row s^n: a0, a2, a4 …; row s^{n-1}: a1, a3, a5 …

3

Next rows

b1 = (a1a2 − a0a3)/a1, b2 = (a1a4 − a0a5)/a1, …

4

Continue

Each element = (pivot × element above-right − element above × pivot-right)/pivot.

Condition: all roots in the LHP ⇔ no sign change in the first column; sign changes = number of RHP roots.

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Example 1: stable

Problem $s^3 + 6s^2 + 11s + 6 = 0$. Is the system stable?

- s^3: 1 11
- s^2: 6 6
- s^1: $(6\times11 - 1\times6)/6 = 10$
- s^0: 6
- First column 1, 6, 10, 6: no sign change

Answer: Stable (roots −1, −2, −3)

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Example 2: unstable

Problem $s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$. How many roots are in the RHP?

- s^4: 1 3 5
- s^3: 2 4
- s^2: $(2\times3 - 1\times4)/2 = 1$, $(2\times5 - 0)/2 = 5$
- s^1: $(1\times4 - 2\times5)/1 = -6$
- s^0: 5. First column 1, 2, 1, −6, 5: two sign changes

Answer: Unstable: 2 roots in the RHP

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Example 3: range of K

Problem Unity feedback, $G = K/[s(s + 1)(s + 2)]$. Find the range of K for stability and the oscillation frequency at the limit.

- Characteristic: $s^3 + 3s^2 + 2s + K = 0$
- s^3: 1 2; s^2: 3 K; s^1: $(6 - K)/3$; s^0: K
- Stable: $(6 - K)/3 > 0$ and $K > 0 \Rightarrow 0 < K < 6$
- $K = 6$: auxiliary $3s^2 + 6 = 0 \Rightarrow s = \pm j\sqrt{2}$

Answer: $0 < K < 6$; marginal oscillation at $\omega = 1.414$ rad/s

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Special case 1: zero in the first column

Problem

A zero appears in the first column with other non-zero elements in the row.

Remedy

Replace 0 by a small $\varepsilon > 0$, continue, then let $\varepsilon \to 0$ and count sign changes.

Example

$s^4 + s^3 + 2s^2 + 2s + 3$: s^2-row first element ε; s^1 element $\to -3/\varepsilon$ (negative): two sign changes $\Rightarrow$ 2 RHP roots.

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Special case 2: a row of zeros

Problem

An entire row is zero: the polynomial has roots symmetric about the origin.

Remedy

Form the auxiliary polynomial A(s) from the row above; replace the zero row by dA/ds coefficients.

Example

$s^3 + 2s^2 + s + 2$: s^1 row zero; $A(s) = 2s^2 + 2 \Rightarrow$ roots $\pm j1$; no sign change $\Rightarrow$ marginally stable.

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Key takeaways

- Routh array: no sign change in the first column ⇒ stable
- Number of sign changes = number of RHP roots
- Zero in first column: use ε; row of zeros: use the auxiliary polynomial
- Range of K from the s^1 and s^0 entries

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Practice questions

Q1. Apply Routh's test to $s^4 + 8s^3 + 18s^2 + 16s + 5$.

Q2. Find K for stability: $s^3 + 5s^2 + 6s + K = 0$.

Q3. Apply Routh to $s^5 + s^4 + 3s^3 + 3s^2 + 2s + 2$ and explain the zero row.

Q4. What does the auxiliary equation represent?